Engaging Students in Developing a Stereoscopic 3D Educational History Game

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Abstract: The aim of this paper is to present an educational history game the creation of which engaged students in the actual developing process of a stereoscopic 3D environment. We believe that designing and using a stereoscopic 3D learning environment is important because it helps students establish a spatial way of thinking. This kind of skill does not appear in most of the curricula subjects, even though evidence appears in the literature that it can be really helpful in our students' school and everyday life. Besides that, this game enhanced our students' IT skills and their knowledge of an ancient Greek era, in terms of the history subject taught in the first year of middle school. The design of the educational game presented is based on the Greek ancient history curriculum as taught in middle-school with a connection to ancient Greek literature, that is with Homer's "Odyssey". The first period designed and examined is the Mycenaean one, but we hope that students will expand the game to other eras of Greek history. The game is about decision making and the ultimate goal is to find the "mistakes" in each era (these are of several types, as in objects, every-day life situations etc.). Finally, the educational value of the specific game is to be found in two levels. First of all, second and third year middle-school as well as high-school students were involved in the design of the game (and thus were the actual creative interactive media producers). In addition, students helped in the process of development of the scenarios to be presented, a fact that allowed them to evaluate their historical knowledge. On the other hand, first year middle-school students were the ones that played the game and checked the knowledge acquired during the specific academic year.

Keywords: educational game, stereoscopic 3D, virtual environments, spatial thinking, students as designers, history subject

1. Do educational computer games help in education?

The evolution of our students' abilities mainly depends on their capability to learn outside classroom. In order to achieve this, their ability to communicate, to judge situations and make decisions, to have a creative and independent way of thinking are only some of the factors needed. Consequently, one of the main goals of the educational process should be to provide students with the opportunity to be independent through strategic thinking, decision making and problem-solving. Educational games have been found to significantly help students advance the skills mentioned above, as well as cognitive skills (Robertson & Nicholson 2007).

In addition, when examining principles of the theory of education, one comes to the conclusion that an "experiential" model of learning helps the education process more than traditional ways. According to the experiential model, teachers create the experience through which students actually learn. The learning outcome is possible because students have been previously given the opportunity to participate in the examination of the actual problem and are involved in finding ways to solve it. The need for experiential learning is one more factor that supports the use of games as an educational tool. Of course, their use is not independent of restrictions, such as a clear educational goal, specific rules or proper design (Kafai 1995).

Moreover, the appeal that computer games have to children has raised the question whether computer games can be used in the educational process. This question has been so far answered in a positive manner in the literature (Prensky 2008) and this is why games-based learning has emerged, along with an effort to create and use computer-based games in education. According to Papert (1993) software games teach children that some forms of learning are fast-paced, compelling and rewarding whereas the traditional school practice strikes many young people as slow and boring.

But even more popular to children nowadays are 3D virtual reality games. Specifically in middle-school, according to Zimmerman et al (2003), curriculum mainly points on the content, which is rather abstract and complex, rather than on the process of learning new knowledge. Complex systems or phenomena, though, are difficult to be understood in a non-3D space; especially those that cannot be dissected in any other way. According to recent research (e.g. Isik-Ercan, Kim & Nowak 2011), 3D visualization can provide motives that change the students' attitude towards a subject and students report that understand even better through stereoscopic 3D visual environments, they learn faster and that the lesson is more "real" and "understandable".

2. Spatial thinking, stereoscopic projection and virtual learning environments

But what skills exactly are empowered in such a stereoscopic 3D learning experience? The actual student's perception of a real system or phenomenon according to the subject's curriculum may be the obvious answer. Is there anything else? In this section we will focus on the spatial thinking as a skill that can be developed in a virtual stereoscopic 3D learning environment.

Spatial cognition includes three elements (Liben as cited in Cohen 1985, p. 6): (1) spatial products are the concrete results of the processing of spatial information (e.g., the spatial objects of historical monument); (2) spatial storage is information stored about space and relationships within space; and (3) spatial thinking is mechanisms and capabilities that help to realize cognitive events in which information processing takes place through interaction with the content of spatial storage and spatial products. Spatial thinking concerns the locations of objects, their shapes, their relations to each other, and the paths they take as they move. We also use spatial thinking to describe non spatial situations, such as when we talk about being close to a goal or describe someone as an insider (Newcombe 2010).

Although spatial thinking is something that everybody should have, school education rarely supports the development of this kind of skills in an organized manner. Because it is not a subject (or a way of learning), spatial thinking is often considered as something extracurricular or abstract (as imagination) (Newcombe 2010). But the truth is that it is required in someone's every day or school life. For example, using a map, giving directions to strangers or exploring a new place require such a way of thinking. Mapping a historical monument, for example an ancient Greek temple, visualizing the way things are located and are useful in its interior or revitalizing the strategy behind a famous battle of Napoleon. Furthermore, spatial thinking includes the emotional exploration of a new friend's behavior and beliefs; this requires an abstract but solid way of finding the limits, linking gaps, building bridges metaphorically, being in someone else's place. In the same way, we believe that exploring the everyday habits, lifestyle and ethics in past civilizations, acknowledging the historical context depends on the existence of spatial thinking.

Research has proved that spatial thinking is a key to the so called STEM (Science, Technology, Engineering and Mathematics) disciplines (Wai et al. 2009). Thus, the fact that spatial thinking is critical for the designers of a 3D stereoscopic learning environment in terms of combining different technologies and knowledge in an interdisciplinary manner for highly visual and close to real representation is more than obvious. We suggest that spatial thinking for the participant, the "user" of a 3D stereoscopic learning environment is actually affected positively when the learning process involves revitalizing spaces, monuments and lifestyle of ancient civilizations. This is mainly because the learner can define better the inner connection among people and between people and objects in a specific historical context. Needless to say that this is presented in a close to real life, a vivid stereo projection.

This kind of 3D stereoscopic projection within a learning environment presents a limited sense of physical presence into the virtual world, when compared to a classic immersive Virtual Reality solution. However, we believe that the fact that the learner retains the sense of being in the classroom while being transferred metaphorically to a stereoscopically projected 3D world under specific circumstances (such as playing a group game after the classic textbook based learning experience) can be something very positive. "Presence is defined as the belief that you are actually in the Virtual Environment and not in the lab or classroom". However, "presence may or may not result from immersion. And presence may be experienced in many ways without using Virtual Reality technology. But, remaining with this simple definition, there is evidence for significant positive correlation between presence and what students learn in a Virtual Environment" (Winn 2005). We would like, as teachers, to retain the dynamics of a class and enhance them with stereoscopic 3D projection. We would also like to consider the learners as a group that interacts with the stereo projected virtual world. To say that they interact implies that the students' perception of participating in a group and their individual sense and knowledge of a new virtual environment are separate things that are coupled (Van Gender 1997). This process definitely involves communication between students, between students and the teacher and between students and the computer through an interface and rules to use it as a team. If some learners (or maybe students that have been previously taught the particular subject) participated in the creation of the virtual world and all students continually gave feedback to the teacher, we would be able to see the learners, the teacher, the stereoscopic virtual world and its student creators as an interconnected whole.

3. Students as designers

Aristotle was one of the ancient Greek philosophers that had spent much time observing the human nature in regards to the ways knowledge is acquired. According to his theory "teaching is the highest form of understanding" and this is why we originally chose to engage students in the actual design of the game. The first question arising is whether students can design games, but this seems to have been positively answered already (Prensky 2008).

Students in order to design games have to evaluate situations, choose an action to be made, judge the results obtained and also get feedback for their decisions. This process is important, since it gives them the opportunity to gain a number of skills, such as designing rules, advancing collaboration and communication with fellow students and teachers and their ability to teach (Sandoval and Bell, 2004). This type of skills can now be found in the literature as "soft skills" and are the ones related to the evolution of the child's emotional intelligence, which is really important for the interaction with others.

Finally, when designing a game, one has to take into account the engagement of the players. Several definitions have been given to the term "engagement", but according to Darren Nonis (2005) they all share the notions of mindfulness, that is paying close attention to something or staying aware of it, cognitive effort and attention. Of course, there are different levels of engagement, ranging from low to high. There are several factors, or "rules of engagement" according to Prensky (2007), that contribute to the success in this domain. One of the most important though, in order to actually have a game, is the factor of fun. It is certainly not easy to define the term, but one can always understand when a game misses it. The lack of fun is one of the main reasons that educational computer games fail to engage the students and, consequently, this leads to failure of the learning process. Quite often educational games created by educators smell too much like school because educators are grownups that cannot fully understand the minds of today's students and the games that they would enjoy playing. So, by getting students to use their after-school computer games experience and activities in designing the game and for the sake of the in-school procedure, the hope is to help students create a fun and engaging game.

4. The process of making the Game

In order to begin the project and get a high level of engagement from our student-designers, we originally asked for volunteers and had quite a large number of responses. According to Prensky (2008), students can be motivated and encouraged to participate with recognition and prizes, but in our case the volunteers, except for the fact that they got involved in a process with materials that they enjoyed, they also had a specific goal: to help in the making of a game that could be tested in the new highly equipped IT laboratory of the school.

The value of using students as designers of the History Game that we describe was really important in several levels. First of all, we had to consider the academic goals of the game. In regards to history, one of the main goals of the teaching of this subject according to the curriculum is for middle and high school students to get acquainted with Greek history and civilization through the eras. In order for the students to get involved in the development of the scenarios and models of the places where facts happened, they had to revise their first-year historical knowledge and interpret the school curriculum. Prerequisite for that was the excellent knowledge of the subject. In addition, getting involved in such a process helped them evolve a positive frame of mind towards the study of the past. This is an assumption to be examined in the future.

The History Game was developed by a group of 8 students aged 14-18, that were second and third year middle school as well as high school students that have been taught about the historic era examined during the first year of middle school. The main concept was that the designers should have the sophistication needed, that was above the level to which the game was targeted, but in addition they had to be not too far removed from their own experience (Prensky 2008). These students combined the IT skills learned in school and after school time, the latter refers to our school's "3D Animation" and "Algorithmic Thinking" clubs. In this way they coupled the curricular and extracurricular activities they participated in. Their necessary IT skills were a combination of 3D Modeling and Design (by using Google Sketchup), 3D Figure Design and Animation (by using DAZ

3D Studio), 3D Stereoscopic Animation (by using iClone 4 Pro and iClone 3DXchange) and Game Programming (by using the Unity3D game engine and programming in Javascript programming language). Almost every application that was used is freeware and ready to download through the internet. Only some educational licenses of iClone Pro were bought and also some 3D outfits for the figures of the Game (DAZ M3 Hoplite outfit models).

In the process of creating the game most students chose to participate in a specific field, but everyone had to observe at least for one hour another student in some stage of the creation of the game, so as to be able to understand and describe the whole project.

Before the creation of the game students that participated in the developing groups were able to:

- Create simple and complex geometrical shapes, create and use textures, use scale, make components, inference, use the offset, move and push/pull tool, use the section tool, use navigation and camera techniques, make copies and arrays, use layers, intersect with models, model accurately doors, windows, walls and roofs (3D Modeling and Design team)
- Use basic lighting and background techniques, create simple characters, use poses, use shadows, use cameras and morphs, create and import texture maps (3D Figure Design and Animation team)
- Use the main interface of the iClone Animation application, operate with cameras, use actors, paths and animation, use the timeline, manage the set and the stage, use the stereoscopic video export feature efficiently (3D Stereoscopic Animation team)
- Use the basic Unity3D engine's functionality and workflow, create simple scripts in Javascript, use the tree creator, use basic interaction techniques, import 3D models and use cameras and lighting efficiently (Game Programming team)

During and especially after the development of the Game all students that participated in the development group were able to:

- Decide the goals, the rules, the characters and the interaction in the designing stage of a game
- Identify the basic stages of a complex game designing and developing
- Explain the use of every 3D object and interaction of the Game
- Design simple workflows and explaining their own choices in terms of complexity
- Decide how to fit their choices to the aesthetics and goals of the Game
- Develop simple stereoscopic walkthroughs
- Apply written storytelling into a virtual world
- Report as a group, giving feedback to others, collaborating with members of different teams
- See the connection between different subjects (History and Computer science) as a unified well-structured process

Of course, an educational game cannot be designed independently of the players. First year middle-school students were the ones that played the game, since the historic era at which the game takes place is being taught the specific year of middle-school.

5. Presentation of the Game

The History Game starts with a virtual stereoscopic walkthrough. At first the user of the Game can experience a small trip back in time. Our school's main building is presented (Figure 1) and the virtual camera slowly turns 180 degrees to the past and the palace of Odysseus (Figure 2). This is used as a topic introduction, as the users sit comfortably and "trip" back in time.

At the same point, a recorded student's voice explains the main goals of our game that are the key elements of the history that is about to be told. The game takes place in the palace of Odysseus, when Goddess Athena –as a man- visits the son of Odysseus, Telemachus, in Odysseus' palace. Several suitors of Penelope –Odysseus wife- are found to be eating and drinking in the palace like they own it. Telemachus accepts the stranger in the palace (Greek hospitality) and complaints about the suitors' behavior (verses 115-173, rhapsody a). Homer's "Odyssey" takes place in the Mycenaean era, as the game itself.

Figure 4: A Penelope's suitor at the stage of designing in DAZ

When the reading stops, it is time for the interactive part of the game to start (this is not yet implemented in stereo vision). Our main character is navigated in and out of the palace with the use of the keyboard and mouse. A student can manage our first-person character according to the instructions of the class (this is a team collaboration activity). If the user clicks on a particular object the system prompts "Is there something wrong with this spot?" (Figure 5)

Figure 5: The system asks whether there is something wrong with the spot

Types of errors include situations (for instance women sitting on a table, something that was not permitted in the particular era) or the existence of 3D objects and buildings that shouldn't be there (for example the existence of an Apollonian temple, a God that wasn't worshiped at the Mycenaean era – Figure 6).

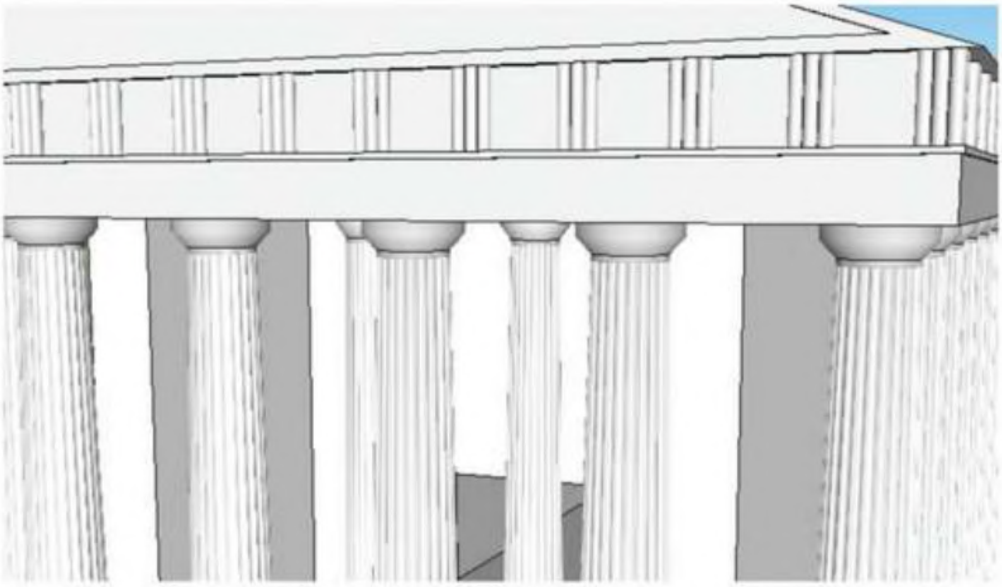

Figure 6: A temple of Apollo exists in the virtual world, an error that has to be found by the user

If every error is found, the time that was spent is being shown to the user. So this is a simple competition against the clock. Two teams can play simultaneously, since our new school lab has three DLP projectors.

The game structure is quite simple, yet can be used for many teamwork activities. However, the next stage is using the Unity3D engine to make it even more fun and intriguing for the users (e.g. add a selection to play as a man or a woman so as to be a part of the cultural context, add network capabilities so as to compete in the same projected environment, add score, more complex situations as errors and more complex menus allowing the user to explain why exactly there is an error in a spot, etc.).

6. Conclusion

According to the above, undoubtedly computer games and specifically stereoscopic 3D ones can act as educational tools, since both the students-designers and the students-players can be benefited by them. Of course, one should always bear in mind that technology is not to be used in the educational process independently of a clear educational goal and without proper design. Students seemed to be engaged in the process and were motivated to learn. In addition, several skills were developed, such as strategic, decision-making, collaboration or problem-solving ones, as well as their spatial thinking was advanced (to what point, however, has to be clarified in our future research). In addition, students cooperated with each other in order to achieve the goal of the game, but they also evaluated the situations presented to them, decided their next move based on facts and tried to solve the problems presented to them. All the above, helped students acquire and evolve skills that are needed in school as well as in their everyday life. Last but not least, in the specific educational history game presented here, there was added value concerning the combined knowledge of the curriculum in two academic areas, IT skills and historic knowledge.

Some of the benefits that our students gained can be proven by the actual result that was achieved and had mainly to do with their involvement in the designing of the game. More specifically, in order to do so, they used 3D model designing, programming knowledge and followed a specific programming context, which was rather complex for their age and, thus, very difficult to teach otherwise. In addition, with the 3D technology provided, they had to imagine and then implement a virtual space retaining the scientific accuracy needed. They had to follow specific rules and serve a general pattern throughout the developing process in a way that any adult developer should. Furthermore, their knowledge about the stereoscopic 3D technology was evolved, since they were already aware of this kind of technology for entertaining purposes, but now they got involved with its educational 'side'.

We believe that even more benefits arose for the students as players. Yet, we didn't have the chance to evaluate them fully, but we plan on doing so in the near future. For example, we expect to measure the Game's positive effect on our players' revision in the subject of history during the specific academic year. Finally, we would like to expand the particular method in terms of its historical context and then clarify and measure key factors that are affected regarding the knowledge of history. In this way, we plan to assess the actual impact on the students' perception of the world they live in as the result of an evolutionary process.

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